الجامعة اللبنانية
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IN410
Cryptography and Secure Communications

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Course Syllabus

- Context
- History/ Classical ciphers
- Symmetric Encryption
- Cryptographic hash functions
- <u>Key Exchange & Public Key Cryptography</u>
- Authentication

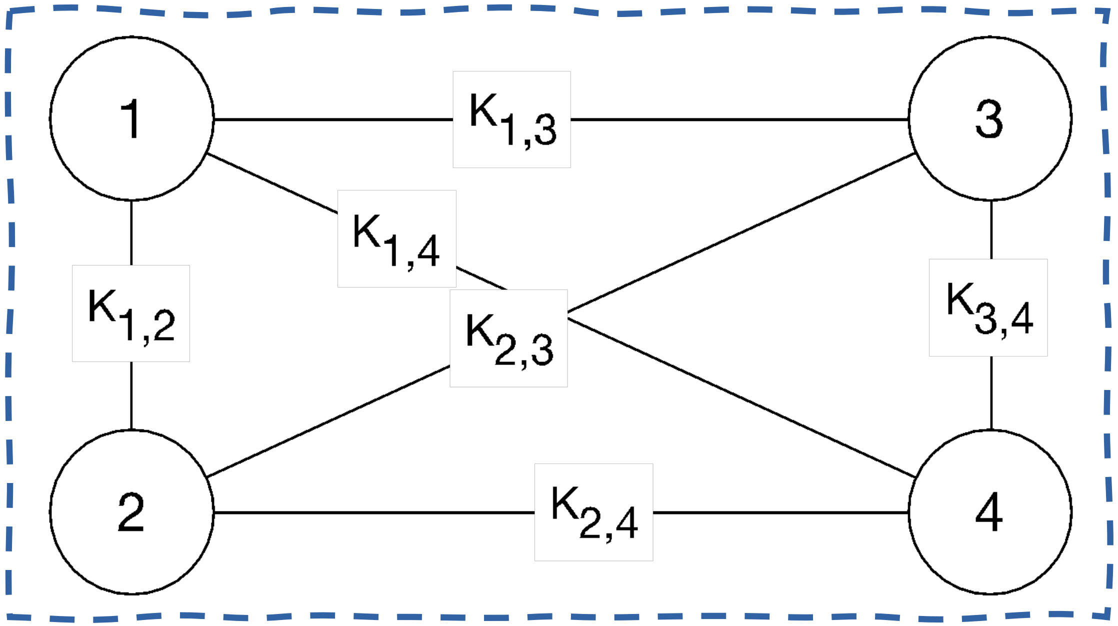
Credits

- The following slides are the adaptation of
  - Lecture slides of Lawrie Brown based on William Stallings book "Cryptography and Network Security: Principles and Practice"
  - Lecture Slides of Dan Boneh – Stanford University
  - Lecture slides of Pawel Wocjan – University of Central California
  - Lecture Slides of Ahmed Serhrouchni – Telecom ParisTech
  - Others (References in the note section)

KEY EXCHANGE

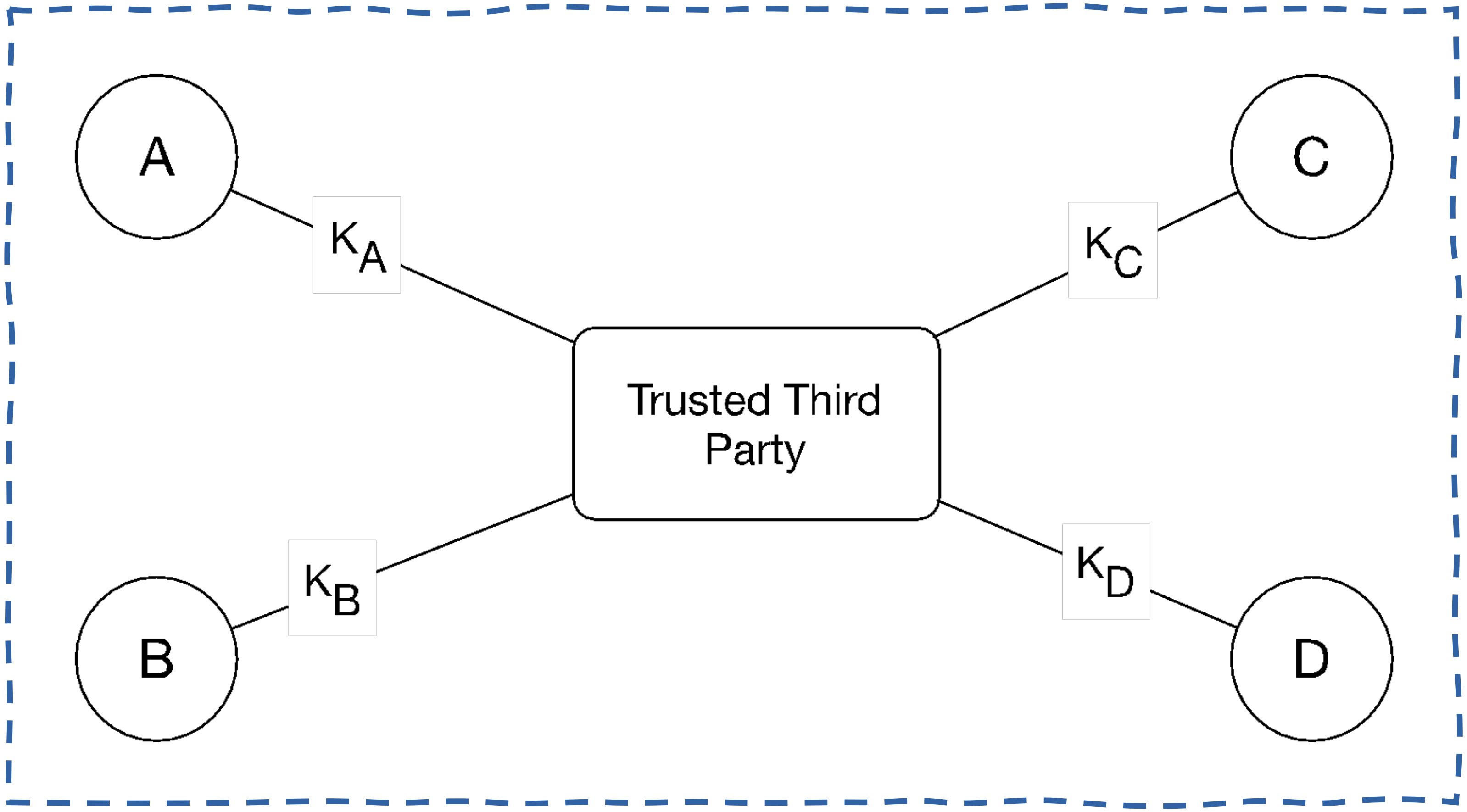
Key Management

- Problem:
  - Storing mutual secret keys is difficult
  - N users => O(N) keys per user



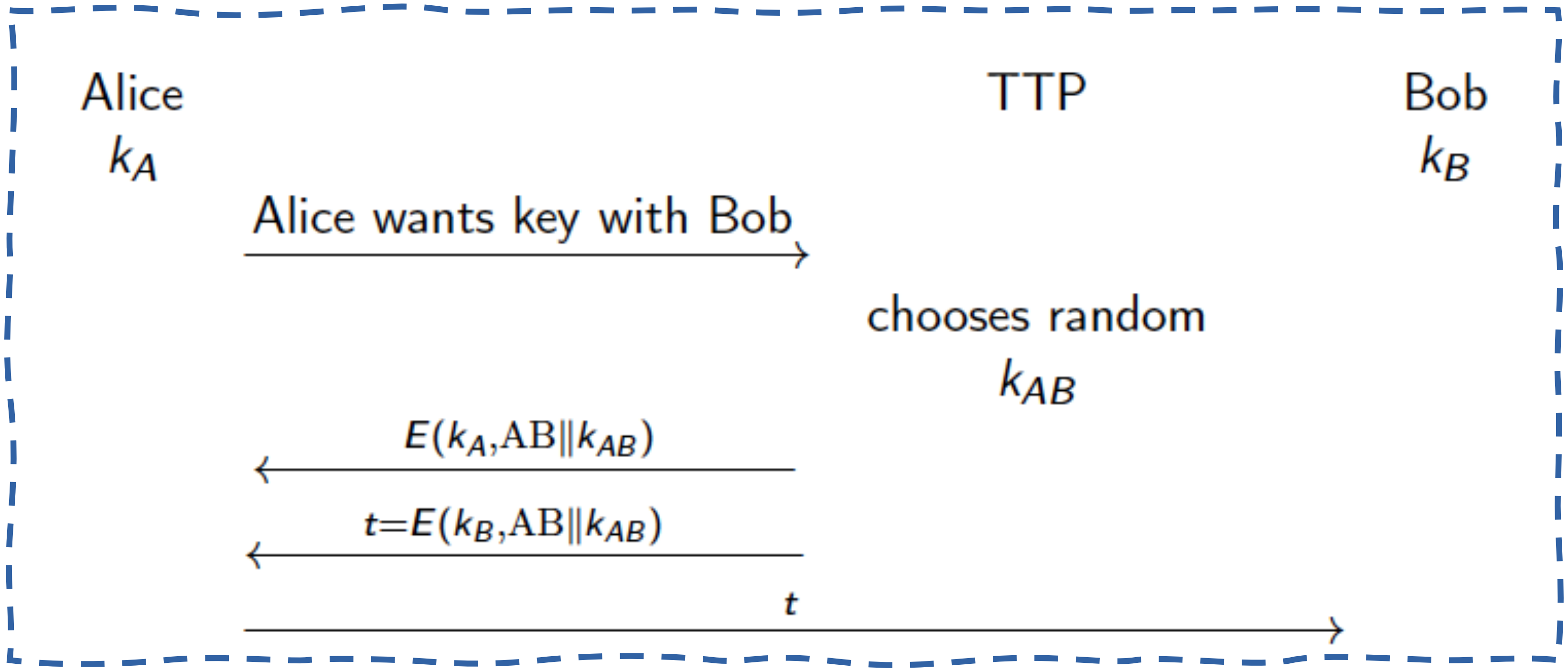
Online Trusted 3rd Party (TTP)

- Every user only remembers one key.



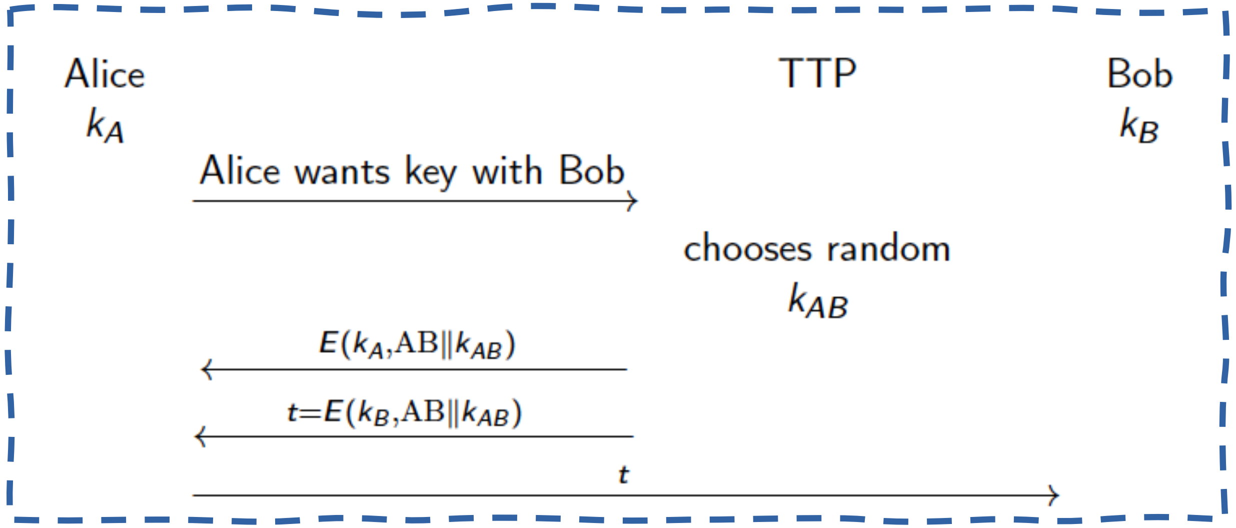
Online Trusted 3rd Party: a toy protocol

- Alice wants a shared key with Bob.



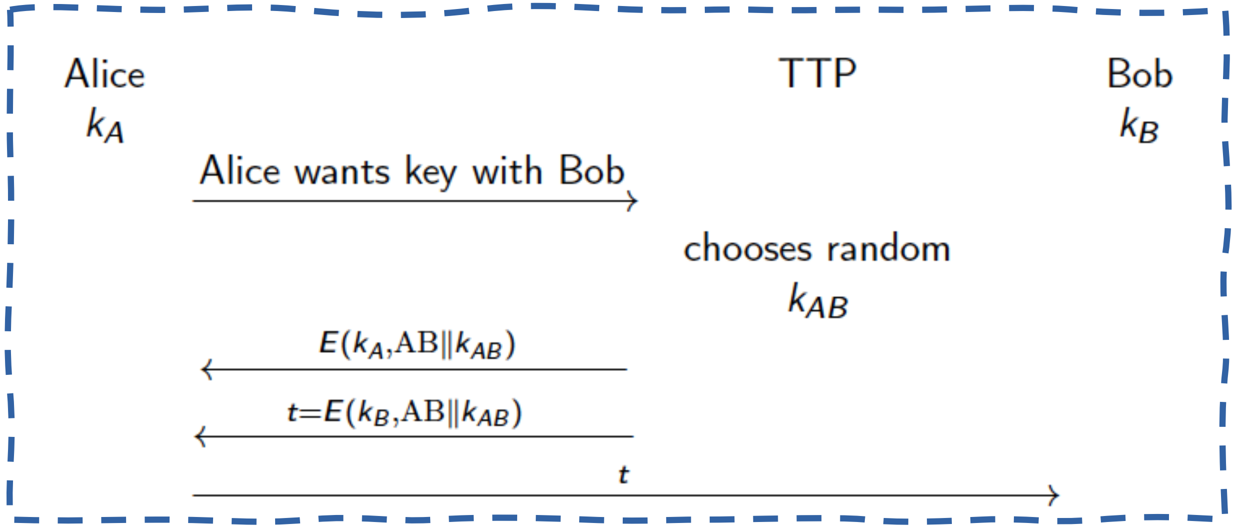
- Eavesdropping security only.

Online Trusted 3rd Party: a toy protocol



- Eavesdropper sees: $E(k_A, \text{“}A, B\text{”} \| k_{AB}); E(k_B, \text{“}A, B\text{”} \| k_{AB})$
- Eavesdropper learns nothing about k_{AB}
- Notes
  - TTP needed for every key exchange
  - knows all session keys
- Basis of the Kerberos System

Online Trusted 3rd Party: a toy protocol



- Insecure against active attacks
  - Example: insecure against replay attacks
    - Attacker records session between Alice and merchant Bob (Example: book order)
    - Attacker replays session to Bob
      - Bob thinks Alice is ordering another copy of book

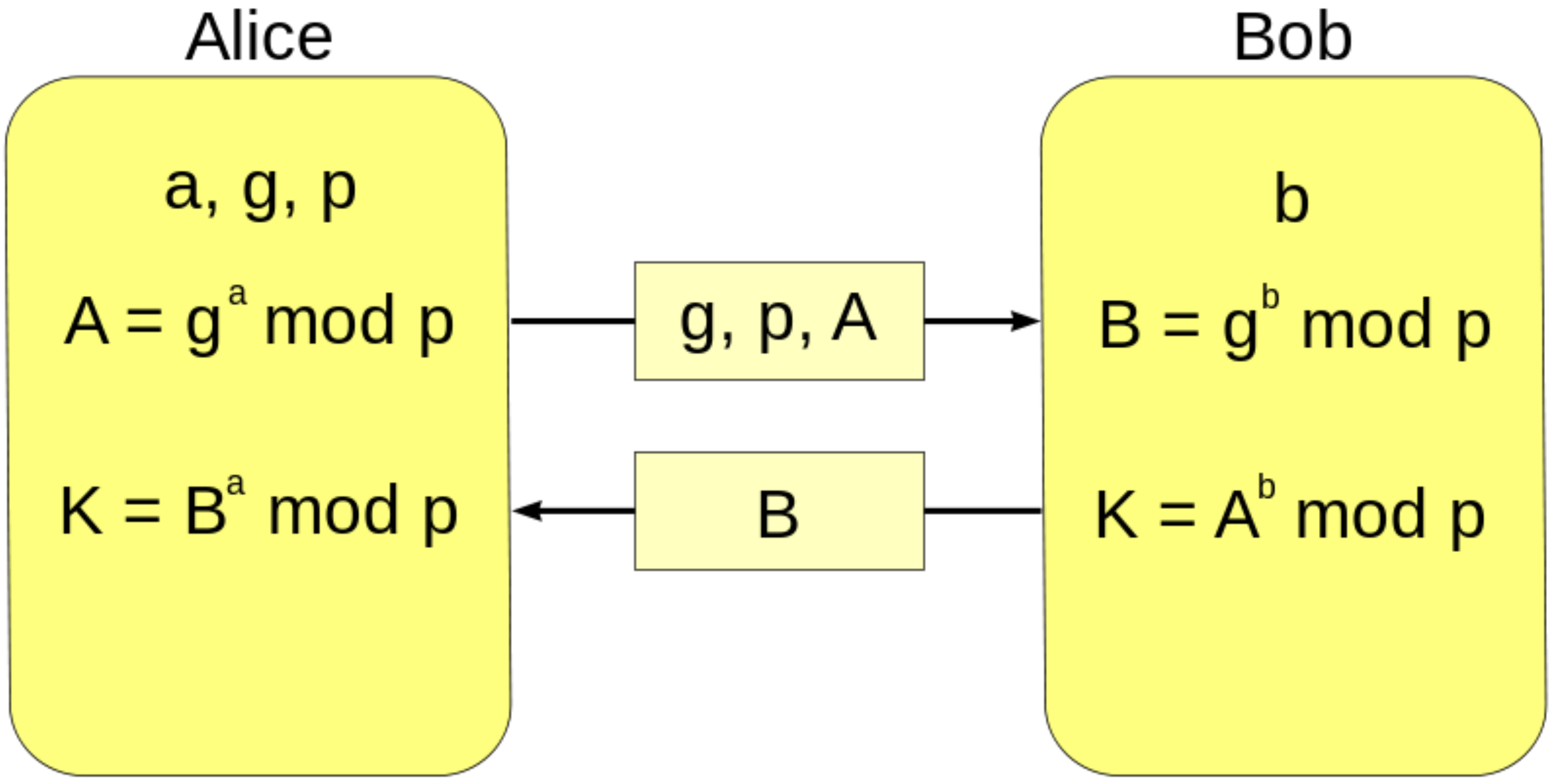
Key question

- Can we generate shared keys without an online trusted 3rd party?
- Starting point of public-key cryptography:
  - Merkle (1974)
  - Diffie-Hellman (1976)
  - RSA (1977)
  - El-Gamal (1985)
  - ID-based encryption
  - Functional encryption
  - …

Diffie-Hellman Key Exchange

- First public-key type scheme
- Proposed by **Diffie & Hellman** in 1976 along with the exposition of public key concepts
- A practical method for public exchange of a secret key
- Based on **exponentiation in a finite field** (Easy)
- Security relies on the difficulty of computing **discrete logarithms** (Hard problem)

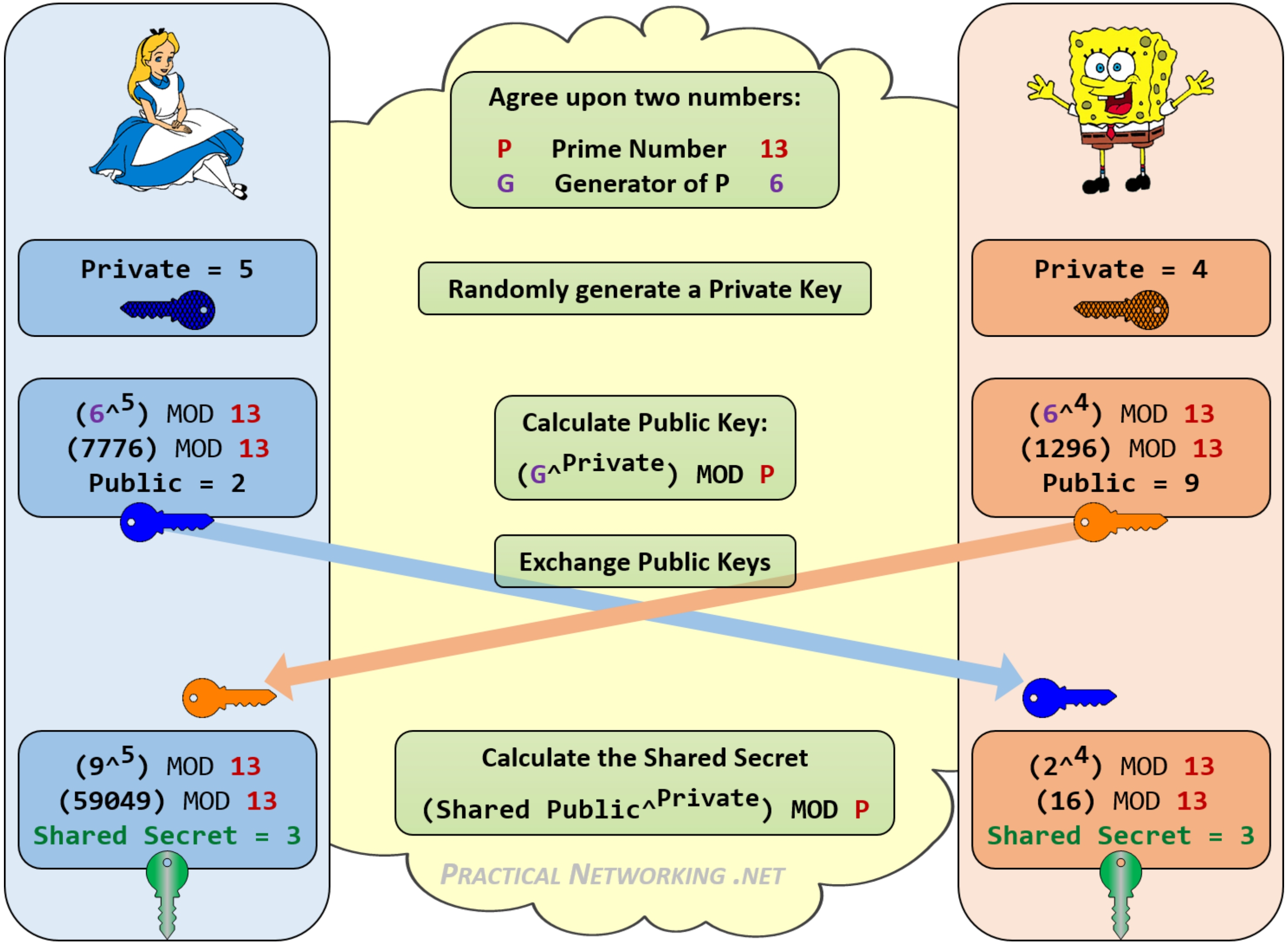
Diffie-Hellman Key Exchange



$$K = A^b \bmod p = (g^a \bmod p)^b \bmod p = g^{ab} \bmod p = (g^b \bmod p)^a \bmod p = B^a \bmod p$$

p: large prime integer or polynomial p
g being a primitive root mod p

Diffie-Hellman Example



Computing Large Modular Exponentiation

- Problem: Compute $a^b mod\ c$ where $b \gg$
- Example:
  - How to compute: $2^{35236} mod\ 35237$
- % is used to represent modulo
  - $a \,\%\, b = a\ mod\ b$

Computing Large Modular Exponentiation

- Reminders
  - $ab\,\%c = [(a\%c)(b\%c)]\%c$
  - $a^b = a^{\frac{b}{2}}.a^{\frac{b}{2}} = (a^{\frac{b}{2}})^2$
    - $a^b\%\,c = [(a^{\frac{b}{2}}\%\,c)(a^{\frac{b}{2}}\%\,c)]\%c$

Computing Large Modular Exponentiation

- **How to compute $a^b \% \, c$**
  - **Write b as a sum of powers of 2 (like in conversion to binary)**
    - Example: 45 = 32 + 8 + 4 + 1
  - **Write a^bas a multiplication of a^{b_i} Where b_i is a multiple of 2**
    - Example: Compute $7^{45}\%45$
      - $7^{45} = 7^{32} * 7^8 * 7^4 * 7^1$
  - **Start computing $a^{b_i}\%c$ starting from the small b_i and by raising to power 2 each time**
    - $7^1\% \ 45 = 7$
    - $7^2\%45 = 7 \ * 7 \ \%45 = 49 \ \%45 = 4$
    - $7^4\%45 = 7^2 * 7^2 \ \%45 = 16 \ \%45 = 16$
    - $7^8\%45 = 7^4 * 7^4 \ \%45 = 16 \ * 16 \ \ \%45 = 256\%45 = 31$
    - $7^{16}\%45 = 7^8 * 7^8 \ \%45 = 31 \ * 31 \ \ \%45 = 961\%45 = 16$
    - $7^{32}\%45 = 7^{16} * 7^{16} \ \%45 = 16 \ * 16 \ \ \%45 = 256\%45 = 31$
  - **Compute a^bas a multiplication of a^{b_i}**
    - $7^{45}\%45 = 7^{32}\%45 \ * 7^8\% \ 45 * 7^4\% \ 45 \ * 7^1\%45 = 7 \ * 16 \ * 31 \ * 31 \ \%45 = 37$

Exercise

- Alice and Bob are using the Diffie-Hellman key exchange protocol to agree on a key for a shift cipher. Suppose the public mod is $n = 31$ and the public base is 3.
  - Alice picks 14 as her secret number. What number does she send to Bob?
  - Bob picks 19 as his secret number. What number does he send to Alice?
  - How does Alice compute the key (K_1) and what does she get?
  - How does Bob compute the key (K_2) and what does he get?

Man-in-the-Middle Attack